

ROADMAP — Major Posteriors NOT in Current Model

Updated 2026-09-12 per R1+R2 review in mapreview.docx. Branch: wip/cloud-9-relhic. Current model: 4D joint posterior ($\log_{\sigma}/m_0$, a , $\log_{\epsilon}$, $\log_{\alpha}$), 10 channels, $\log_Z = -2.94$ (TIER-3 WEAKLY CONSISTENT). NOTE: 10^{-54} means 10 to the -54 power.

Summary table — 7 missing posteriors, ranked (with reviewer priority bumps)

#	Missing posterior	Dimensions	New total	Priority	Cost
1	DM mass (m_{χ})	+1	5D	HIGH	1-2 weeks
2	Mediator mass ($m_{A'}$)	+1	5D	HIGH	1-2 weeks
1+2	$m_{\chi} + m_{A'}$ jointly	+2	6D	HIGH	2-4 weeks
4	Baryonic feedback	structural	—	HIGH (R1)	4-8 weeks OR 1-2 weeks minimal
6	Cross-channel correlations	structural	—	MEDIUM (R1)	1-2 weeks OR 1 week minimal
3	Particle-physics model class	structural	—	MEDIUM	2-4 weeks
7	Time-dependence (gravothermal)	+1	5D	LOW (defer)	2-4 weeks
5	Cosmological initial conditions	structural	—	LOW (defer)	3-6 months

NEW: Phase 0 — Theoretical Assessment of ϵ Naturalness (1-2 weeks)

Per R1 (mapreview.docx Gap 1, paragraph 28): 'Adding m_{χ} and $m_{A'}$ will not answer [the naturalness question]. Cross-channel correlations will not answer this. Baryonic feedback will not answer this. The naturalness problem is a model-building problem, not a fitting problem.'

Tasks: (1) Survey known suppression mechanisms (spontaneously broken $U(1)$, clockwork, large extra dimensions, sequestering). (2) For each: can it produce $\epsilon \sim 10^{-54}$ without new fine-tuning? (3) Decision: if ≥ 1 works $\rightarrow$ proceed with Phases 2-5; if NO mechanism works $\rightarrow$ pivot to Option B (publish null result).

Verdict softened (R1): TIER-3 RESOLVED $\rightarrow$ WEAKLY CONSISTENT

Per Reviewer 1 (roadmap1.docx, paragraph 11): $\log_Z = -2.94$ corresponds to 'substantial' evidence on the Jeffreys scale, not 'strong'. The earlier verdict 'TIER-3 RESOLVED' was overclaiming.

Revised verdict: TIER-3 WEAKLY CONSISTENT with multi-channel data. The marginalization succeeds ($\log_Z = -2.94$ vs catastrophic T30/T32), but the absolute evidence is weak per Jeffreys scale.

Three honest framings of the same posterior

1. Strongest claim: 'Tier-3 marginalization restores consistency with multi-channel data.'

2. Weakest claim: 'Tier-3 marginalization succeeds only if the mediator is decoupled from the Standard Model at the 10^{-54} level — a naturalness catastrophe.'

3. Most honest claim: 'The model is consistent with data ONLY if the SIDM mediator is invisible to the Standard Model. That conditional is the entire result, not a caveat.'

STOP RULE — Do not add more channels (R2, paragraph 32)

Reviewer 2's critical warning: 'Do not expand the number of exotic channels (extra LRD variants, more UDG subtypes, etc.) until the naturalness issue is under better control.'

Channel 11 (DM-free UDG): pre-existing, KEEP. Channel 13 (DM-dominated UDG): added this session, KEEP. Any further channels (Ch14+) should be DEFERRED until: (1) ϵ naturalness addressed, (2) cross-channel correlations fixed, (3) $m_\chi + m_A$ added.

Kill criteria per phase (NEW per R1 Gap 2)

Per R1 (paragraph 38): 'A credible scientific program needs a falsification threshold. ... A roadmap without a kill criterion is a plan to keep going regardless of what the data says.'

Phase	Kill criterion	Action if triggered
Phase 0 (theoretical)	No known mechanism naturally produces $\epsilon \sim 10^{-54}$	Pivot to Option B (publish null result)
Phase 2 (correlations)	log Z degrades below -10 after adding shared nuisance	Stop — model not credible
Phase 3 (baryonic)	CDM + feedback reproduces all 10 channels equally well	SIDM interpretation not required
Phase 4 ($m_\chi + m_A$)	6D log Z worse than 4D log Z (after ΔAIC)	Extra params not justified
Phase 5 (mediator class)	Bayes factors between classes indistinguishable ($ \Delta\log Z < 1$)	Mediator class not constrained
Phase 6 (time-dep)	AMUSE $N \geq 10^4$ simulations show no qualitative change	Time-dep negligible, drop
Phase 7 (alternative DM)	If we got this far, model survived all earlier phases	Pursue as new project

Minimal alternatives (NEW per R2)

R2 (paragraph 120): 'Full hydro + SIDM comparison is a research-group-scale effort. A minimal one- or two-parameter nuisance (e.g., a simple core-size or feedback-efficiency parameter on the SPARC/UDG channels) is achievable faster and still addresses the fundamental degeneracy flagged by the reviewer.'

R2 (paragraph 133): 'Implement a minimal cross-channel correlation treatment — 1 week maximum. ... Do not let it become a multi-week blocker.'

Cross-channel correlations: 1 week minimal (R2) vs 1-2 weeks full (R1). Baryonic feedback: 1-2 weeks minimal (R2) vs 4-8 weeks full (R1). Recommended path: run minimal first.

Null result as legitimate outcome (R1, paragraphs 64-70)

Before Phase 2 begins, decide which outcome to pursue:

(A) Continue with Phase 2+: assume the model is worth pursuing (default)

(B) Pivot to publishing the null result: publish the naturalness catastrophe as the contribution

(C) Pivot to the symmetry question: investigate what symmetry could enforce $\epsilon \sim 10^{-54}$

(D) Defer to particle-physics collaboration: hand off the ϵ problem to model-builders

Revised sequence (per R1 paragraph 73)

1. Phase 0 (theoretical, 1-2 weeks) — first, NEW
2. Phase 2 (correlations, 1-2 weeks) — prerequisite, not parallel
3. Phase 3 (baryonic feedback, 4-8 weeks or 1-2 minimal) — break SIDM/feedback degeneracy
4. Phase 4 ($m_\chi + m_{A'}$ jointly, 2-4 weeks) — only after Phases 0, 2, 3 complete
5. Phase 5 (mediator class, 2-4 weeks) — last
6. Phases 6-7 (defer indefinitely per R1) — these are refinements to a model that may not survive Phase 0

Quick stats

- Total estimated cost: ~3-6 months for particle-physics-grade model.
- Quickest impact: Phase 0 (1-2 weeks) + Phase 2 (1-2 weeks) = ~1 month of preparation.
- Highest risk: Phase 0 (no mechanism may work) → pivot to null result.
- Source doc: v0.3-prelim/docs/ROADMAP_MISSING_POSTERiors_2026_09_12.md (605 lines).
- Reviewers: R1 (paragraphs 0-88, mapreview.docx), R2 (paragraphs 90-138, mapreview.docx).

What changed from previous version (a61eebb)

- Phase 0 ADDED (theoretical assessment of ϵ naturalness) per R1 Gap 1
- Dimensional accounting FIXED: $m_\chi + m_{A'}$ together = 6D, not 5D (R1 Gap 3)
- Kill criteria ADDED per phase (R1 Gap 2)
- Sequence REORDERED: Phase 0 first, Phase 2 (correlations) before Phase 4 (masses)
- Minimal alternatives ADDED (1-2 param baryonic, 1-week correlation) per R2
- Phase 7 deferred indefinitely per R1 (model may not survive Phase 0)
- Superscript notation: unicode → ASCII 10^{-54} (failure-proof rendering in PDF + Telegram)